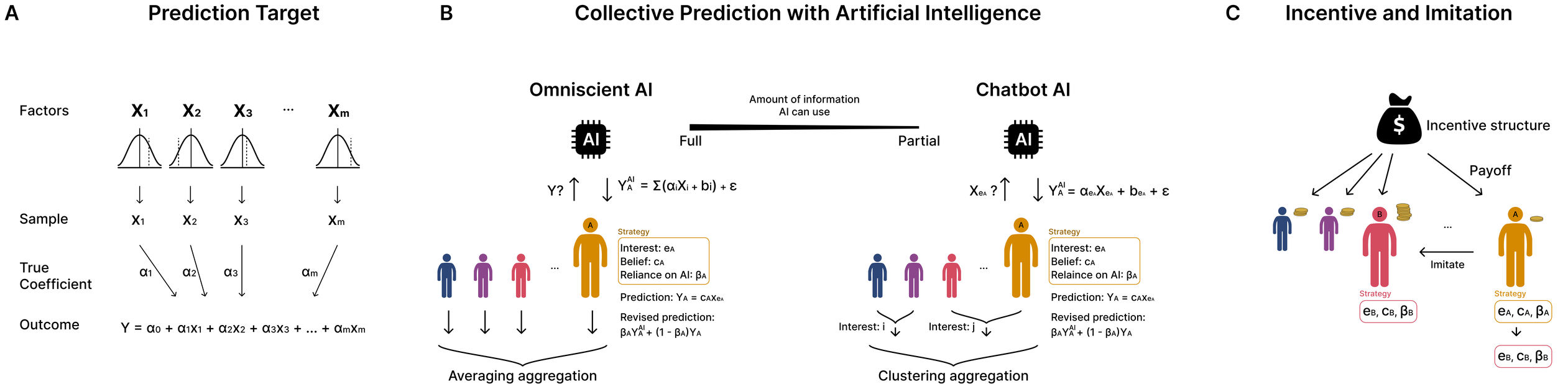
A
Prediction Target
Factors
X1 X2 X3 ... Xm
Sample
X1 X2 X3 Xm
True Coefficient
α1 α2 α3 αm
Outcome
Y = α0 + α1X1 + α2X2 + α3X3 + ... + αmXm
B
Collective Prediction with Artificial Intelligence
Omniscient AI
Amount of information AI can use
Full
Partial
Chatbot AI
AI
Y? ↑ ↓ $Y_A^{AI} = \Sigma(\alpha_i X_i + b_i) + \varepsilon$
X_{e_A} ? ↑ ↓ $Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \varepsilon$
A
Strategy
Interest: e_A
Belief: c_A
Reliance on AI: β_A
Relaince on AI: β_A
Prediction: $Y_A = c_A x_{e_A}$
Revised prediction: $\beta_A Y_A^{AI} + (1 - \beta_A) Y_A$
Interest: i
Interest: j
Averaging aggregation
Clustering aggregation
C
Incentive and Imitation
$
Incentive structure
Payoff
B
Imitate
Strategy
e_B, c_B, β_B
e_A, c_A, β_A